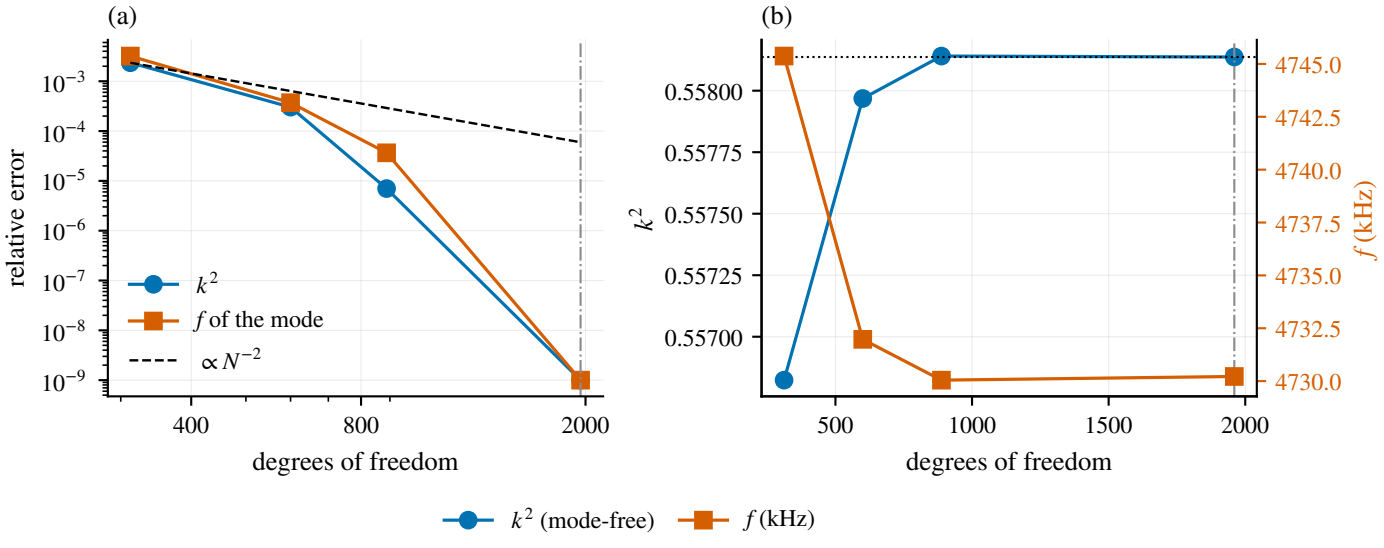
(a)
relative error
$10^{-3}$
$10^{-4}$
$10^{-5}$
$10^{-6}$
$10^{-7}$
$10^{-8}$
$10^{-9}$
$k^2$
$f$ of the mode
$\propto N^{-2}$
400
800
2000
degrees of freedom
(b)
$k^2$
0.55800
0.55775
0.55750
0.55725
0.55700
$f$ (kHz)
4745.0
4742.5
4740.0
4737.5
4735.0
4732.5
4730.0
500
1000
1500
2000
degrees of freedom
$k^2$ (mode-free)
$f$ (kHz)